

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of B. Pharm. Examination

University Theory Examination January-2018

BPH1004T PHARMACEUTICAL INORGANIC CHEMISTRY

Date: 03/01/2018, Wednesday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 75

Instructions:

1. There are three sections in this question paper.
2. **SECTION – I** comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to **SECTION – I** are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for **SECTION – I** is 30 minutes. Answers to **SECTION – I** must be written during the first 30 minutes of the examination.
3. Answers to **SECTION – II** and **SECTION – III** are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark.

20

- 1 If a solution is buffered at a pH of 8.0, then the addition of a small amount of acid will cause the solution to have a pH of approximately
[A] 6.0
[B] 7.0
[C] 8.0
[D] 9.0
- 2 If the pH of a solution increases from 4.0 to 6.0, the hydronium ion concentration
[A] increase by a factor of 1.5
[B] increase by a factor of 100
[C] decrease by a factor of 10
[D] decrease by a factor of 100
- 3 Excess use of magnesium sulphate leads to
[A] hypermagnesaemia
[B] gastrointestinal irritations
[C] watery diarrhoea
[D] all of the above
- 4 In the limit test for chloride, standard solution is
[A] 0.0824 % W/V NaCl solution
[B] 5.0824 % w/v NaCl Solution
[C] 1.824 % w/v NaCl Solution
[D] 1.00 N HCl
- 5 One of the following has a major role in transmission of nerve impulse
[A] Sodium
[B] Magnesium
[C] Calcium
[D] None

- 6 In limit test of sulphate which of following use to prevent supersaturation ?
 [A] Potassium sulphate
 [B] Barium Sulphate
 [C] Alcohol
 [D] None
- 7 New formula Oral rehydration salts contain
 [A] Sodium Chloride, Glucose anhydrous, Potassium Chloride and Trisodium citrate
 [B] Sodium Chloride, Glucose anhydrous, Potassium Chloride
 [C] Sodium Chloride, Glucose anhydrous, Potassium Chloride and Trisodium tartrate
 [D] Sodium Chloride, Glucose anhydrous
- 8 Describe the term for solvent from 10 to 30 parts of the solute
 [A] Very soluble
 [B] Freely soluble
 [C] soluble
 [D] Sparingly soluble
- 9 Zinc-Eugenol Cement Consist of EXCEPT
 [A] Zinc Acetate
 [B] Zinc Sulphate
 [C] Zinc Oxide
 [D] Zinc Stearate
- 10 Fluoride inhibits caries formation via
 [A] decrease acid solubility of enamel
 [B] bacterial inhibition
 [C] increase acid solubility of enamel
 [D] Both A & B
- 11 Which of the following are units are used for measurement of radioactivity?
 [A] Curie
 [B] Roentgen
 [C] Electron volts
 [D] All of the above
- 12 Calamine is a mixture of
 [A] Calcium oxide and zinc oxide
 [B] Calcium oxide and ferric oxide
 [C] Zinc oxide and ferric oxide
 [D] None of the above
- 13 Calculate pH of solution in which the H^+ concentration is $4.2 \times 10^{-4} \text{ mol dm}^{-3}$
 [A] 3.39
 [B] 3.38
 [C] 3.5
 [D] 4
- 14 One of the following is sedative expectorant
 [A] NH_4Cl
 [B] Anise
 [C] Lemon

[D] Eucylaptus

15 Emetic out through local irritation or gastric mucosa

[A] Ipecacaunaha

[B] Ammonium bicarbonate

[C] Ipecacaunaha and Ammonium bicarbonate Both

[D] None

16 Intracellular fluid constituent

[A] 30-50% of body weight

[B] 45-50% of body weight

[C] 12-15% of body wight

[D] 4-5 % of body wight

17 One of the following is well in cyanide poisoning

[A] Sodium nitrate

[B] Sodium Thiosulfate

[C] Sodium nitrate and Sodium Thiosulfate Both

[D] None of above

18 Antidotes act by producing the effect oppose to that of poison is known as

[A] Physiological antidotes

[B] Chemical antidotes

[C] Mechanical antidotes

[D] All

19 Radiopharmaceuticals use in diagnosis of pernicioal anaemia

[A] Ca-45

[B] Cr-51

[C] Co-57

[D] K-42

20 Concentration of fluoride required for use as anticaries agents agent orally is

[A] 5%

[B] 6%

[C] 2%

[D] 4%

SECTION – II

- Q 2 Discuss in details limit test for Chloride and Iron. 10
- Q 3 Attempt any TWO of the following:
- A Discuss methods of preparation, assay and uses of Calcium Gluconate. 05
- B Write a short note on Dentifrices. 05
- C Enlist various methods for measurement of radio activity. Explain Geiger Muller Counter in detail. 05

SECTION – III

- Q 4 Attempt any ONE of the following:
- A What is antacids? State ideal requirements of antacids. Discuss methods of preparation, assay and uses of Sodium Bicarbonate. 10
- B Write a note on pharmacopoeia and mention contents of official monograph. 10
- Q 5 Attempt any FIVE of the following:
- A State the Pharmaceutical Applications of radioactive substances. 05
- B State mechanism of antimicrobial agents. 05
- C Enumerate sources of impurities in pharmaceutical substances. Explain any one in detail. 05
- D What are emetics? Outline the method of preparation, uses and assay of Copper Sulphate. 05
- E Define and classify antidote with examples. 05
- F Define the terms: (1) Astringent, (2) Haematinics, (3) Expectorant, (4) Bacteriostatic and (5) Cathartics 05